Understanding Backpropagation Through a Recurrent Neural Network

A worked example with two layers and two time steps

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When reading a sentence, the meaning of each word depends partly on the words that came before it. A recurrent neural network, or RNN, also carries information between time steps through a hidden state. This lets it use earlier inputs when processing the current input. RNNs are used for ordered data such as text, speech and measurements collected over time [1, 5].

During forward propagation, the network combines the current input with the previous hidden state to calculate a new hidden state. It produces a prediction, and a loss function measures its error. How does that error tell the network which weights to change?

Backpropagation through time (BPTT) traces the loss backwards [2]. With basic matrix multiplication and the chain rule, follow the forward pass, local derivatives, parameter gradients and batch update for one sample.

Illustrative example

This illustrative example is inspired by species-abundance data available in GMrepo v3, a human gut microbiome database [4]. The teaching task is ASD versus non-ASD classification, with class 1 representing ASD and class 0 representing non-ASD.

The example uses two illustrative species abundances from one sample, one at each time step. Here, time steps identify the order in which the abundances enter the RNN; they are not biological measurements taken at different times. The numbers used below are not measured observations or evidence of classification performance.

Two-layer RNN architecture

The model uses a stacked vanilla RNN with two basic recurrent layers, two time steps and two hidden units in each layer. Layer 2 receives the corresponding hidden state from layer 1. Only its final hidden state produces a prediction, making this a many-to-one architecture.

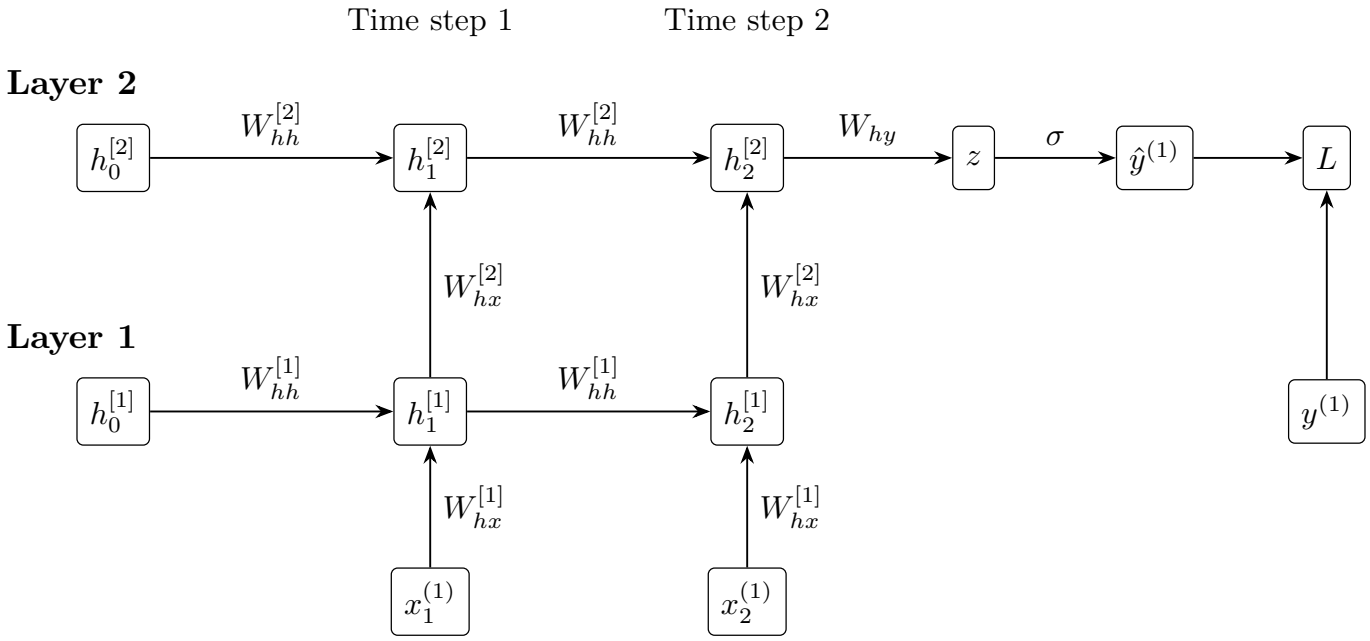


Fig. 1. Two-layer RNN architecture.

Glossary

Notation	Meaning in this example
x_t	Input abundance at time step t. Each input in layer 1 is a scalar, so its input size is 1.
$x_1^{(1)}, x_2^{(1)}$	The two species abundances from sample 1. Together they form a sequence of length 2.
t; subscripts 0, 1, 2	Time-step index. The subscript 0 denotes the initial state; 1 and 2 denote the two time steps.
Superscript (1)	Round brackets distinguish a sample number from a layer number.
Superscripts [1], [2]	Layer 1 and layer 2. Square brackets identify the layer.
$h_0^{[1]}, h_0^{[2]}$	Initial hidden states. Both are zero vectors with shape 2×1.
$h_t^{[1]}, h_t^{[2]}$	Hidden states at time step t. The hidden size is 2, so each state has shape 2×1.
$W_{hx}^{[1]}, W_{hx}^{[2]}$	Input-weight matrices. Their shapes are 2×1 for layer 1 and 2×2 for layer 2.
$W_{hh}^{[1]}, W_{hh}^{[2]}$	Recurrent-weight matrices that transform the previous hidden state. Both have shape 2×2.
$\beta^{[1]}, \beta^{[2]}$	Bias vectors for the two recurrent layers. Each has shape 2×1.
ReLU	Rectified linear unit. The activation replaces each negative entry with zero and preserves each positive entry.
W_{hy}	Output-weight vector, stored as a 2×1 column in this example.
σ	Sigmoid activation, which converts the logit into a value between 0 and 1.
$\hat{y}$	Predicted probability of class 1, which represents ASD in this illustrative task.
y	True class label: 1 for ASD and 0 for non-ASD.
L	Binary cross-entropy loss for one sample. It measures the error in the predicted probability.
$\partial L/\partial W$	The gradient of loss with respect to a weight matrix W. Its shape matches W.
T	Transpose: exchanges rows and columns. It arranges the vectors and matrices into the required shapes.
η	Learning rate: controls the size of a parameter update.
B	Mini-batch size: the number of samples used to calculate one averaged gradient.
$\theta; \theta_{\text{old}}, \theta_{\text{new}}$	Any trainable weight or bias; its value before and after an update.
L_{batch}	Mean loss across the samples in a mini-batch.
$L^{(s)}; s$	Loss for sample s; s identifies a sample within the mini-batch.

Table 1. Notation used in the example.

Graph notation. In each recurrent block, a is the input-weight contribution, b is the previous-state contribution, c is their sum with the layer bias, and h is the ReLU output. The logit z enters sigmoid; q is the scalar output bias. The sample superscript is omitted below because only sample 1 is used.

Reading the graph. Follow the forward arrows to the loss, then work backwards. Combine each local derivative with its incoming gradient. Add contributions where paths meet and across uses of shared weights. Figures 3–6 show the four recurrent blocks of Fig. 2 separately.

Sample 1 and initialized parameters

Species A: $x_1^{(1)} = 0.20$. Species B: $x_2^{(1)} = 0.30$. One abundance enters at each step. These are illustrative values.

Initial hidden states and biases — all zero

$$h_0^{[1]} = h_0^{[2]} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}, \qquad \beta^{[1]} = \beta^{[2]} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}, \qquad q = 0.$$

Here $\beta^{[1]}$ and $\beta^{[2]}$ are the layer biases; q is the output bias. The illustrative target is $y^{(1)} = 1$.

Layer 1 input weights — 2 × 1

$$W_{hx}^{[1]} = \begin{bmatrix} .25 \\ 1.00 \end{bmatrix}.$$

Layer 1 recurrent weights — 2 × 2

$$W_{hh}^{[1]} = \begin{bmatrix} .75 & .50 \\ .50 & 1.00 \end{bmatrix}.$$

Layer 2 input weights — 2 × 2

$$W_{hx}^{[2]} = \begin{bmatrix} .25 & .75 \\ .25 & .25 \end{bmatrix}.$$

Layer 2 recurrent weights — 2 × 2

$$W_{hh}^{[2]} = \begin{bmatrix} .75 & 1.00 \\ .75 & 1.00 \end{bmatrix}.$$

Output weights — 2 × 1

$$W_{hy} = \begin{bmatrix} .50 \\ .50 \end{bmatrix}, \qquad W_{hy}^T = \begin{bmatrix} .50 & .50 \end{bmatrix}.$$

Forward pass through layer 1

Formulas for both hidden states

$$h_1^{[1]} = \mathrm{ReLU}(W_{hx}^{[1]} x_1 + W_{hh}^{[1]} h_0^{[1]} + \beta^{[1]}).$$
$$h_2^{[1]} = \mathrm{ReLU}(W_{hx}^{[1]} x_2 + W_{hh}^{[1]} h_1^{[1]} + \beta^{[1]}).$$

Substitute and solve

$$h_1^{[1]} = \mathrm{ReLU}\left(\begin{bmatrix} .25 \\ 1 \end{bmatrix}(0.20) + \begin{bmatrix} .75 & .5 \\ .5 & 1 \end{bmatrix}\begin{bmatrix} 0 \\ 0 \end{bmatrix} + \begin{bmatrix} 0 \\ 0 \end{bmatrix}\right)$$
$$= \mathrm{ReLU}\left(\begin{bmatrix} .05 \\ .2 \end{bmatrix} + \begin{bmatrix} 0 \\ 0 \end{bmatrix} + \begin{bmatrix} 0 \\ 0 \end{bmatrix}\right) \quad = \mathrm{ReLU}\left(\begin{bmatrix} .05 \\ .2 \end{bmatrix}\right) = \begin{bmatrix} .05 \\ .2 \end{bmatrix}.$$
$$h_2^{[1]} = \mathrm{ReLU}\left(\begin{bmatrix} .25 \\ 1 \end{bmatrix}(0.30) + \begin{bmatrix} .75 & .5 \\ .5 & 1 \end{bmatrix}\begin{bmatrix} .05 \\ .2 \end{bmatrix} + \begin{bmatrix} 0 \\ 0 \end{bmatrix}\right)$$
$$= \mathrm{ReLU}\left(\begin{bmatrix} .075 \\ .3 \end{bmatrix} + \begin{bmatrix} .1375 \\ .225 \end{bmatrix} + \begin{bmatrix} 0 \\ 0 \end{bmatrix}\right) \quad = \mathrm{ReLU}\left(\begin{bmatrix} .2125 \\ .525 \end{bmatrix}\right) = \begin{bmatrix} .2125 \\ .525 \end{bmatrix}.$$

Forward pass through layer 2

Formulas for both hidden states

$h_1^{[2]} = \text{ReLU}(W_{hx}^{[2]} h_1^{[1]} + W_{hh}^{[2]} h_0^{[2]} + \beta^{[2]})$.

$h_2^{[2]} = \text{ReLU}(W_{hx}^{[2]} h_2^{[1]} + W_{hh}^{[2]} h_1^{[2]} + \beta^{[2]})$.

Substitute and solve

$$h_1^{[2]} = \text{ReLU}(\begin{bmatrix} .25 & .75 \\ .25 & .25 \end{bmatrix} \begin{bmatrix} .05 \\ .2 \end{bmatrix} + \begin{bmatrix} .75 & 1 \\ .75 & 1 \end{bmatrix} \begin{bmatrix} 0 \\ 0 \end{bmatrix} + \begin{bmatrix} 0 \\ 0 \end{bmatrix})$$

$$= \text{ReLU}(\begin{bmatrix} .1625 \\ .0625 \end{bmatrix} + \begin{bmatrix} 0 \\ 0 \end{bmatrix} + \begin{bmatrix} 0 \\ 0 \end{bmatrix}) \quad = \text{ReLU}(\begin{bmatrix} .1625 \\ .0625 \end{bmatrix}) = \begin{bmatrix} .1625 \\ .0625 \end{bmatrix}.$$

$$h_2^{[2]} = \text{ReLU}(\begin{bmatrix} .25 & .75 \\ .25 & .25 \end{bmatrix} \begin{bmatrix} .2125 \\ .525 \end{bmatrix} + \begin{bmatrix} .75 & 1 \\ .75 & 1 \end{bmatrix} \begin{bmatrix} .1625 \\ .0625 \end{bmatrix} + \begin{bmatrix} 0 \\ 0 \end{bmatrix})$$

$$= \text{ReLU}(\begin{bmatrix} .446875 \\ .184375 \end{bmatrix} + \begin{bmatrix} .184375 \\ .184375 \end{bmatrix} + \begin{bmatrix} 0 \\ 0 \end{bmatrix}) \quad = \text{ReLU}(\begin{bmatrix} .63125 \\ .36875 \end{bmatrix}) = \begin{bmatrix} .63125 \\ .36875 \end{bmatrix}.$$

Output prediction

$$z = W_{hy}^T h_2^{[2]} + q, \qquad \hat{y} = \sigma(z) = \frac{1}{1 + e^{-z}}.$$

$$z = \begin{bmatrix} .5 & .5 \end{bmatrix} \begin{bmatrix} .63125 \\ .36875 \end{bmatrix} + 0 = (0.50)(0.63125) + (0.50)(0.36875) + 0 = 0.50.$$

$$\hat{y} = \frac{1}{1 + e^{-0.50}} = 0.622459331.$$

The predicted probability of class 1 (ASD) is approximately 0.62246 for this illustrative sample.

Prediction loss

Binary cross-entropy compares the predicted probability with the true binary label. This loss is calculated once, after the final time step.

$$L = -[y \ln(\hat{y}) + (1 - y) \ln(1 - \hat{y})].$$

For the illustrative ASD sample, the target is y = 1:

$$L = -[1 \ln(0.622459331) + (1 - 1) \ln(1 - 0.622459331)]$$
$$= -[1(-0.474076984) + 0(-0.974076984)] = 0.474076984.$$

Backpropagation Part 1 Calculating Local Derivative

A computational graph records the operations that connect the inputs and parameters to the loss. A local derivative describes how the output of one operation changes when its input changes. The incoming gradient, also called the global gradient here, describes how that output affects the final loss. The chain rule combines the incoming gradient with the local derivative. At a branch, contributions from all routes are added.

Backpropagation computational graph

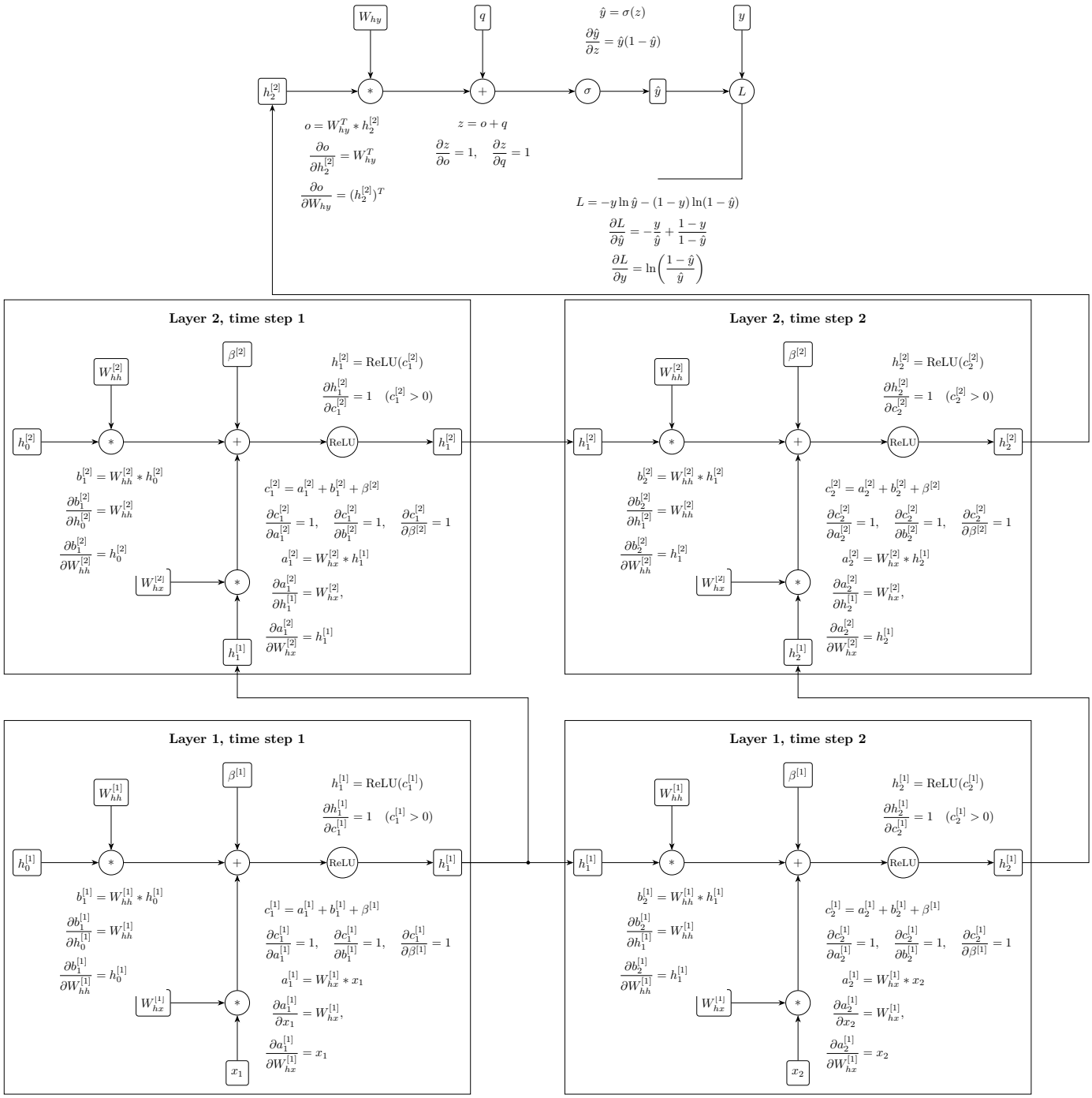


Fig. 2. Complete computational graph with local derivatives. Larger panels are shown in Figs. 3–7.

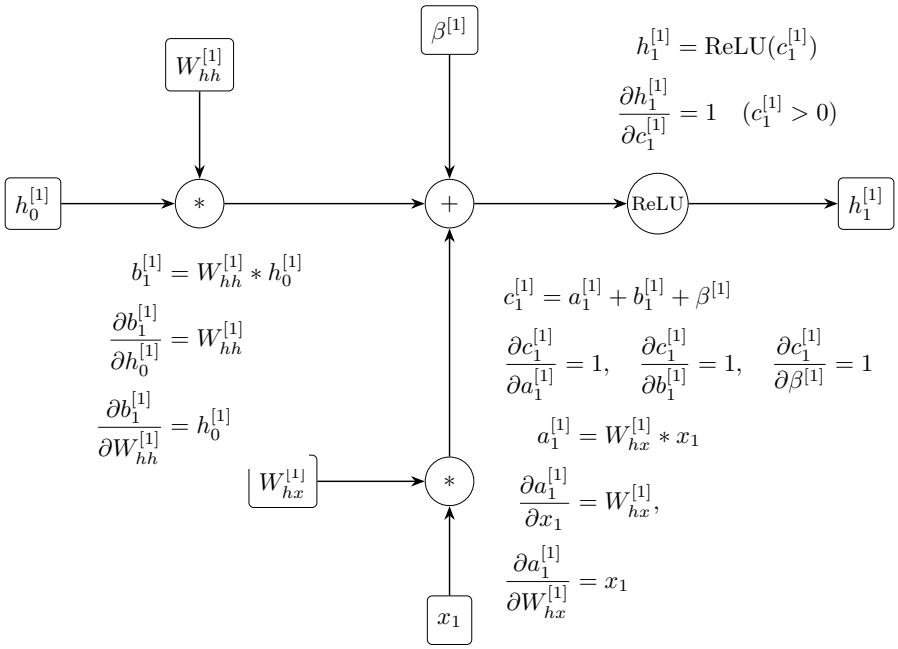


Fig. 3. Computational graph for layer 1 at time step 1.

Local derivatives: layer 1, time step 1

Node a — input multiplication

$$a_1^{[1]} = W_{hx}^{[1]} x_1 = \begin{bmatrix} .05 \\ .2 \end{bmatrix}.$$

$$\frac{\partial a_1^{[1]}}{\partial x_1} = W_{hx}^{[1]} = \begin{bmatrix} .25 \\ 1 \end{bmatrix}. \qquad \frac{\partial a_1^{[1]}}{\partial (W_{hx}^{[1]})} = x_1 = 0.2.$$

Node b — recurrent multiplication

$$b_1^{[1]} = W_{hh}^{[1]} h_0^{[1]} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}.$$

$$\frac{\partial b_1^{[1]}}{\partial h_0^{[1]}} = W_{hh}^{[1]} = \begin{bmatrix} .75 & .5 \\ .5 & 1 \end{bmatrix}. \qquad \frac{\partial b_1^{[1]}}{\partial (W_{hh}^{[1]})} = h_0^{[1]} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}.$$

Node c — addition with the bias

$$c_1^{[1]} = a_1^{[1]} + b_1^{[1]} + \beta^{[1]} = \begin{bmatrix} .05 \\ .2 \end{bmatrix}.$$

$$\frac{\partial c_1^{[1]}}{\partial a_1^{[1]}} = 1, \qquad \frac{\partial c_1^{[1]}}{\partial b_1^{[1]}} = 1, \qquad \frac{\partial c_1^{[1]}}{\partial \beta^{[1]}} = 1.$$

Node h — ReLU

$$h_1^{[1]} = \mathrm{ReLU}(c_1^{[1]}) = \begin{bmatrix} .05 \\ .2 \end{bmatrix}. \qquad \frac{\partial h_1^{[1]}}{\partial c_1^{[1]}} = \mathrm{ReLU}'(c_1^{[1]}) = 1.$$

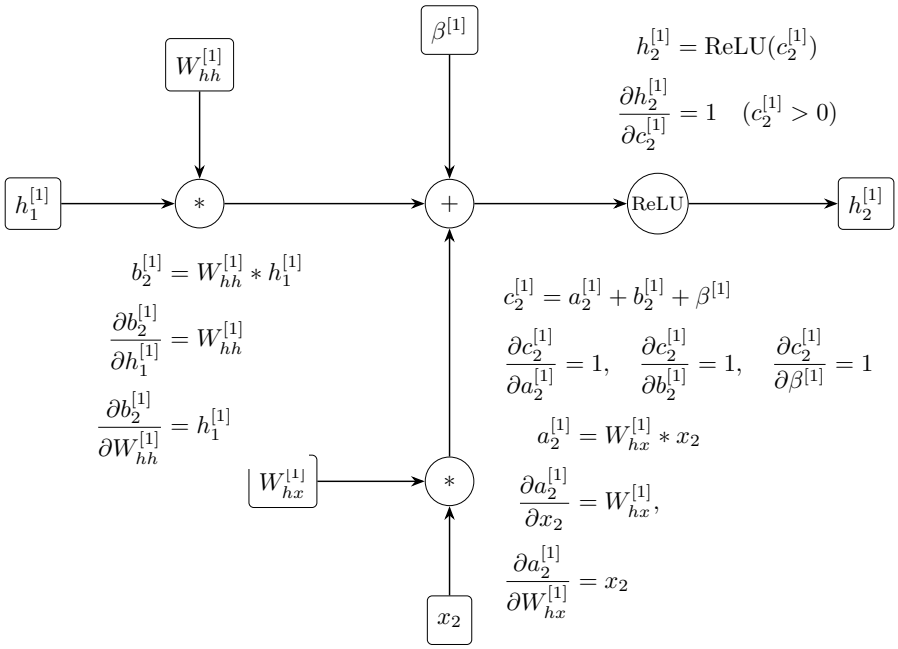


Fig. 4. Computational graph for layer 1 at time step 2.

Local derivatives: layer 1, time step 2

Node a — input multiplication

$$a_2^{[1]} = W_{hx}^{[1]} x_2 = \begin{bmatrix} .075 \\ .3 \end{bmatrix}.$$

$$\frac{\partial a_2^{[1]}}{\partial x_2} = W_{hx}^{[1]} = \begin{bmatrix} .25 \\ 1 \end{bmatrix}. \qquad \frac{\partial a_2^{[1]}}{\partial (W_{hx}^{[1]})} = x_2 = 0.3.$$

Node b — recurrent multiplication

$$b_2^{[1]} = W_{hh}^{[1]} h_1^{[1]} = \begin{bmatrix} .1375 \\ .225 \end{bmatrix}.$$

$$\frac{\partial b_2^{[1]}}{\partial h_1^{[1]}} = W_{hh}^{[1]} = \begin{bmatrix} .75 & .5 \\ .5 & 1 \end{bmatrix}. \qquad \frac{\partial b_2^{[1]}}{\partial (W_{hh}^{[1]})} = h_1^{[1]} = \begin{bmatrix} .05 \\ .2 \end{bmatrix}.$$

Node c — addition with the bias

$$c_2^{[1]} = a_2^{[1]} + b_2^{[1]} + \beta^{[1]} = \begin{bmatrix} .2125 \\ .525 \end{bmatrix}.$$

$$\frac{\partial c_2^{[1]}}{\partial a_2^{[1]}} = 1, \qquad \frac{\partial c_2^{[1]}}{\partial b_2^{[1]}} = 1, \qquad \frac{\partial c_2^{[1]}}{\partial \beta^{[1]}} = 1.$$

Node h — ReLU

$$h_2^{[1]} = \mathrm{ReLU}(c_2^{[1]}) = \begin{bmatrix} .2125 \\ .525 \end{bmatrix}. \qquad \frac{\partial h_2^{[1]}}{\partial c_2^{[1]}} = \mathrm{ReLU}'(c_2^{[1]}) = 1.$$

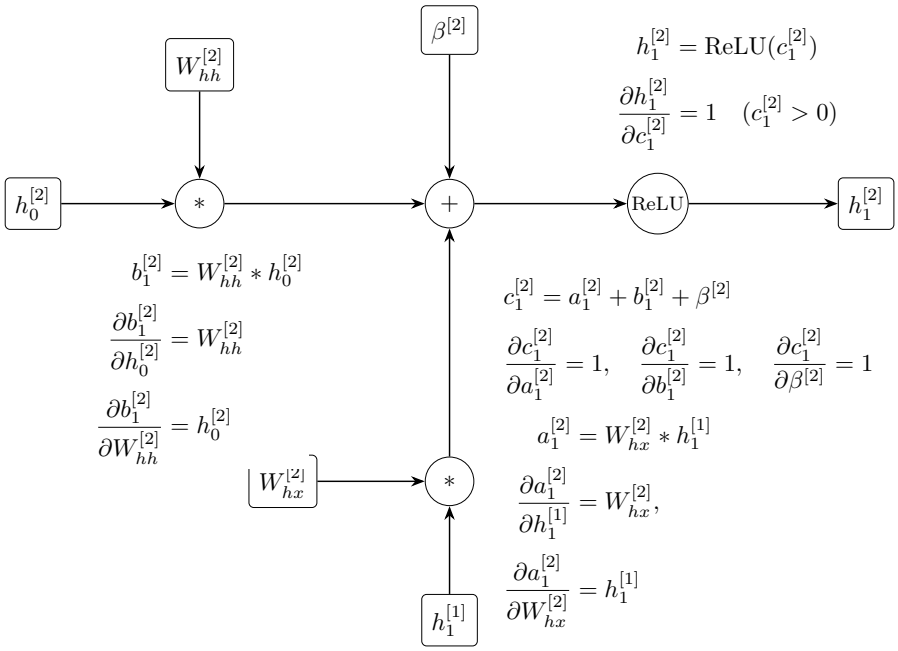


Fig. 5. Computational graph for layer 2 at time step 1.

Local derivatives: layer 2, time step 1

Node a — input multiplication

$$a_1^{[2]} = W_{hx}^{[2]} h_1^{[1]} = \begin{bmatrix} .1625 \\ .0625 \end{bmatrix}.$$

$$\frac{\partial a_1^{[2]}}{\partial h_1^{[1]}} = W_{hx}^{[2]} = \begin{bmatrix} .25 & .75 \\ .25 & .25 \end{bmatrix}. \qquad \frac{\partial a_1^{[2]}}{\partial (W_{hx}^{[2]})} = h_1^{[1]} = \begin{bmatrix} .05 \\ .2 \end{bmatrix}.$$

Node b — recurrent multiplication

$$b_1^{[2]} = W_{hh}^{[2]} h_0^{[2]} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}.$$

$$\frac{\partial b_1^{[2]}}{\partial h_0^{[2]}} = W_{hh}^{[2]} = \begin{bmatrix} .75 & 1 \\ .75 & 1 \end{bmatrix}. \qquad \frac{\partial b_1^{[2]}}{\partial (W_{hh}^{[2]})} = h_0^{[2]} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}.$$

Node c — addition with the bias

$$c_1^{[2]} = a_1^{[2]} + b_1^{[2]} + \beta^{[2]} = \begin{bmatrix} .1625 \\ .0625 \end{bmatrix}.$$

$$\frac{\partial c_1^{[2]}}{\partial a_1^{[2]}} = 1, \qquad \frac{\partial c_1^{[2]}}{\partial b_1^{[2]}} = 1, \qquad \frac{\partial c_1^{[2]}}{\partial \beta^{[2]}} = 1.$$

Node h — ReLU

$$h_1^{[2]} = \mathrm{ReLU}(c_1^{[2]}) = \begin{bmatrix} .1625 \\ .0625 \end{bmatrix}. \qquad \frac{\partial h_1^{[2]}}{\partial c_1^{[2]}} = \mathrm{ReLU}'(c_1^{[2]}) = 1.$$

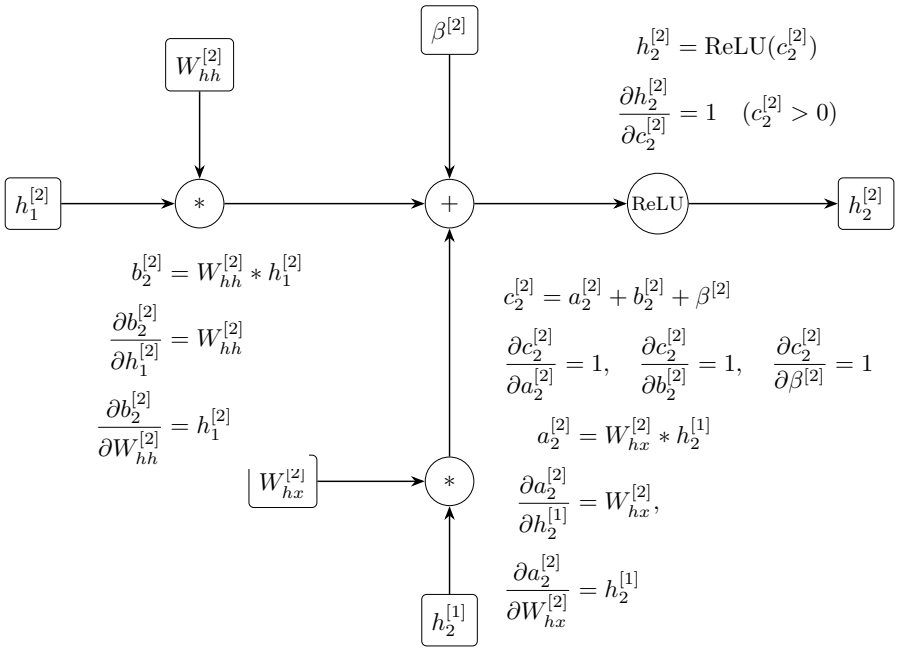


Fig. 6. Computational graph for layer 2 at time step 2.

Local derivatives: layer 2, time step 2

Node a — input multiplication

$$a_2^{[2]} = W_{hx}^{[2]} h_2^{[1]} = \begin{bmatrix} .446875 \\ .184375 \end{bmatrix}.$$

$$\frac{\partial a_2^{[2]}}{\partial h_2^{[1]}} = W_{hx}^{[2]} = \begin{bmatrix} .25 & .75 \\ .25 & .25 \end{bmatrix}. \qquad \frac{\partial a_2^{[2]}}{\partial (W_{hx}^{[2]})} = h_2^{[1]} = \begin{bmatrix} .2125 \\ .525 \end{bmatrix}.$$

Node b — recurrent multiplication

$$b_2^{[2]} = W_{hh}^{[2]} h_1^{[2]} = \begin{bmatrix} .184375 \\ .184375 \end{bmatrix}.$$

$$\frac{\partial b_2^{[2]}}{\partial h_1^{[2]}} = W_{hh}^{[2]} = \begin{bmatrix} .75 & 1 \\ .75 & 1 \end{bmatrix}. \qquad \frac{\partial b_2^{[2]}}{\partial (W_{hh}^{[2]})} = h_1^{[2]} = \begin{bmatrix} .1625 \\ .0625 \end{bmatrix}.$$

Node c — addition with the bias

$$c_2^{[2]} = a_2^{[2]} + b_2^{[2]} + \beta^{[2]} = \begin{bmatrix} .63125 \\ .36875 \end{bmatrix}.$$

$$\frac{\partial c_2^{[2]}}{\partial a_2^{[2]}} = 1, \qquad \frac{\partial c_2^{[2]}}{\partial b_2^{[2]}} = 1, \qquad \frac{\partial c_2^{[2]}}{\partial \beta^{[2]}} = 1.$$

Node h — ReLU

$$h_2^{[2]} = \mathrm{ReLU}(c_2^{[2]}) = \begin{bmatrix} .63125 \\ .36875 \end{bmatrix}. \qquad \frac{\partial h_2^{[2]}}{\partial c_2^{[2]}} = \mathrm{ReLU}'(c_2^{[2]}) = 1.$$

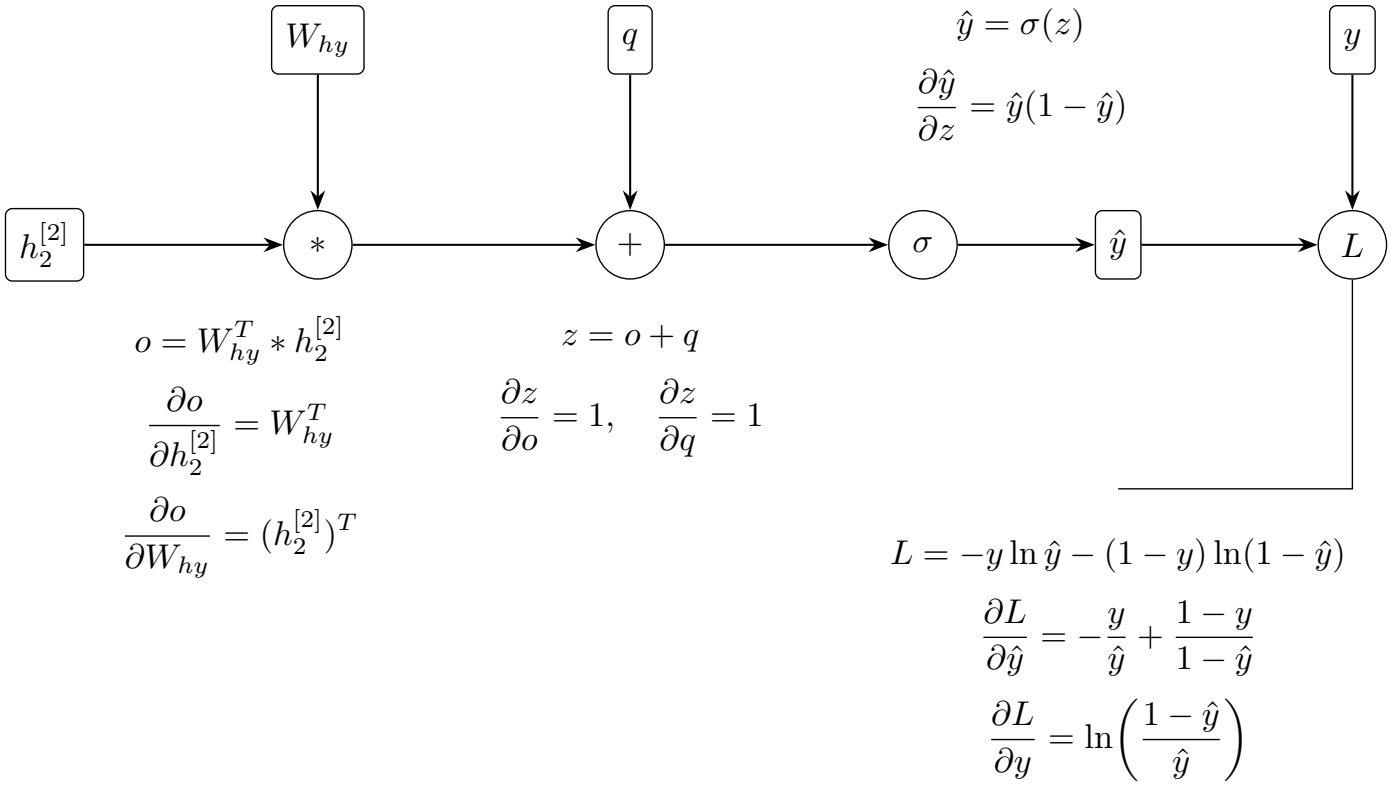


Fig. 7. Output and loss computational graph.

Local derivatives: output and loss

$o = W_{hy}^T h_2^{[2]} = 0.50, \qquad z = o + q = 0.50.$

$\frac{\partial o}{\partial h_2^{[2]}} = W_{hy}^T = \begin{bmatrix} .5 & .5 \end{bmatrix}.$

$\frac{\partial o}{\partial W_{hy}} = (h_2^{[2]})^T = \begin{bmatrix} .63125 & .36875 \end{bmatrix}.$

$\frac{\partial z}{\partial o} = 1, \qquad \frac{\partial z}{\partial q} = 1.$

$\hat{y} = \sigma(z) = 0.622459331, \qquad \frac{\partial \hat{y}}{\partial z} = \hat{y}(1-\hat{y}) = 0.235003712.$

$L = -y \ln \hat{y} - (1-y)\ln(1-\hat{y}) = 0.474076984.$

$\frac{\partial L}{\partial \hat{y}} = -\frac{y}{\hat{y}} + \frac{1-y}{1-\hat{y}} = -1.606530660.$

$\frac{\partial L}{\partial y} = -\ln \hat{y} + \ln(1-\hat{y}) = -0.50.$

The target and abundance inputs are fixed; only the weights and biases are updated.

Backpropagation Part 2 Calculating the gradients

Layer 1 recurrent weight gradients

$$\frac{\partial L}{\partial W_{hh}^{[1]}} = [\frac{\partial L}{\partial \hat{y}}\frac{\partial \hat{y}}{\partial z}\frac{\partial z}{\partial h_2^{[2]}}\frac{\partial h_2^{[2]}}{\partial c_2^{[2]}} \times (\frac{\partial c_2^{[2]}}{\partial h_1^{[2]}}\frac{\partial h_1^{[2]}}{\partial c_1^{[2]}}\frac{\partial c_1^{[2]}}{\partial h_1^{[1]}} + \frac{\partial c_2^{[2]}}{\partial h_2^{[1]}}\frac{\partial h_2^{[1]}}{\partial c_2^{[1]}}\frac{\partial c_2^{[1]}}{\partial h_1^{[1]}})\frac{\partial h_1^{[1]}}{\partial c_1^{[1]}}]^T (h_0^{[1]})^T + [\frac{\partial L}{\partial \hat{y}}\frac{\partial \hat{y}}{\partial z}\frac{\partial z}{\partial h_2^{[2]}}\frac{\partial h_2^{[2]}}{\partial c_2^{[2]}}\frac{\partial c_2^{[2]}}{\partial h_2^{[1]}}\frac{\partial h_2^{[1]}}{\partial c_2^{[1]}}]^T (h_1^{[1]})^T. \tag{1}$$

Numerical substitution

$$\frac{\partial L}{\partial W_{hh}^{[1]}} = [(-1.60653066)(0.235003712)\begin{bmatrix}.5 & .5\end{bmatrix}\begin{bmatrix}1 & 0\\ 0 & 1\end{bmatrix} \times (\begin{bmatrix}.75 & 1\\ .75 & 1\end{bmatrix}\begin{bmatrix}1 & 0\\ 0 & 1\end{bmatrix}\begin{bmatrix}.25 & .75\\ .25 & .25\end{bmatrix} + \begin{bmatrix}.25 & .75\\ .25 & .25\end{bmatrix}\begin{bmatrix}1 & 0\\ 0 & 1\end{bmatrix}\begin{bmatrix}.75 & .5\\ .5 & 1\end{bmatrix})\begin{bmatrix}1 & 0\\ 0 & 1\end{bmatrix}]^T\begin{bmatrix}0 & 0\end{bmatrix}$$

$$+ [(-1.60653066)(0.235003712)\begin{bmatrix}.5 & .5\end{bmatrix}\begin{bmatrix}1 & 0\\ 0 & 1\end{bmatrix}\begin{bmatrix}.25 & .75\\ .25 & .25\end{bmatrix}\begin{bmatrix}1 & 0\\ 0 & 1\end{bmatrix}]^T\begin{bmatrix}.05 & .2\end{bmatrix}.$$

Result

$$\frac{\partial L}{\partial W_{hh}^{[1]}} = \begin{bmatrix}-0.004719258 & -0.018877033\\ -0.009438517 & -0.037754067\end{bmatrix}.$$

Layer 2 recurrent weight gradients

$$\frac{\partial L}{\partial W_{hh}^{[2]}} = [\frac{\partial L}{\partial \hat{y}}\frac{\partial \hat{y}}{\partial z}\frac{\partial z}{\partial h_2^{[2]}}\frac{\partial h_2^{[2]}}{\partial c_2^{[2]}} \times \frac{\partial c_2^{[2]}}{\partial h_1^{[2]}}\frac{\partial h_1^{[2]}}{\partial c_1^{[2]}}]^T (h_0^{[2]})^T + [\frac{\partial L}{\partial \hat{y}}\frac{\partial \hat{y}}{\partial z}\frac{\partial z}{\partial h_2^{[2]}}\frac{\partial h_2^{[2]}}{\partial c_2^{[2]}}]^T (h_1^{[2]})^T. \tag{2}$$

Numerical substitution

$$\frac{\partial L}{\partial W_{hh}^{[2]}} = [(-1.60653066)(0.235003712)\begin{bmatrix}.5 & .5\end{bmatrix}\begin{bmatrix}1 & 0\\ 0 & 1\end{bmatrix} \times \begin{bmatrix}.75 & 1\\ .75 & 1\end{bmatrix}\begin{bmatrix}1 & 0\\ 0 & 1\end{bmatrix}]^T\begin{bmatrix}0 & 0\end{bmatrix}$$

$$+ [(-1.60653066)(0.235003712)\begin{bmatrix}.5 & .5\end{bmatrix}\begin{bmatrix}1 & 0\\ 0 & 1\end{bmatrix}]^T\begin{bmatrix}.1625 & .0625\end{bmatrix}.$$

Result

$$\frac{\partial L}{\partial W_{hh}^{[2]}} = \begin{bmatrix}-0.030675179 & -0.011798146\\ -0.030675179 & -0.011798146\end{bmatrix}.$$

Layer 1 input weight gradients

$$\frac{\partial L}{\partial W_{hx}^{[1]}} = [\frac{\partial L}{\partial \hat{y}}\frac{\partial \hat{y}}{\partial z}\frac{\partial z}{\partial h_2^{[2]}}\frac{\partial h_2^{[2]}}{\partial c_2^{[2]}} \times (\frac{\partial c_2^{[2]}}{\partial h_1^{[2]}}\frac{\partial h_1^{[2]}}{\partial c_1^{[2]}}\frac{\partial c_1^{[2]}}{\partial h_1^{[1]}} + \frac{\partial c_2^{[2]}}{\partial h_2^{[1]}}\frac{\partial h_2^{[1]}}{\partial c_2^{[1]}}\frac{\partial c_2^{[1]}}{\partial h_1^{[1]}})\frac{\partial h_1^{[1]}}{\partial c_1^{[1]}}]^T x_1 + [\frac{\partial L}{\partial \hat{y}}\frac{\partial \hat{y}}{\partial z}\frac{\partial z}{\partial h_2^{[2]}}\frac{\partial h_2^{[2]}}{\partial c_2^{[2]}}\frac{\partial c_2^{[2]}}{\partial h_2^{[1]}}\frac{\partial h_2^{[1]}}{\partial c_2^{[1]}}]^T x_2. \tag{3}$$

Numerical substitution

$$\frac{\partial L}{\partial W_{hx}^{[1]}} = [(-1.60653066)(0.235003712)\begin{bmatrix}.5 & .5\end{bmatrix}\begin{bmatrix}1 & 0\\ 0 & 1\end{bmatrix} \times (\begin{bmatrix}.75 & 1\\ .75 & 1\end{bmatrix}\begin{bmatrix}1 & 0\\ 0 & 1\end{bmatrix}\begin{bmatrix}.25 & .75\\ .25 & .25\end{bmatrix} + \begin{bmatrix}.25 & .75\\ .25 & .25\end{bmatrix}\begin{bmatrix}1 & 0\\ 0 & 1\end{bmatrix}\begin{bmatrix}.75 & .5\\ .5 & 1\end{bmatrix})\begin{bmatrix}1 & 0\\ 0 & 1\end{bmatrix}]^T(0.2) + [(-1.60653066)(0.235003712)\begin{bmatrix}.5 & .5\end{bmatrix}\begin{bmatrix}1 & 0\\ 0 & 1\end{bmatrix}\begin{bmatrix}.25 & .75\\ .25 & .25\end{bmatrix}\begin{bmatrix}1 & 0\\ 0 & 1\end{bmatrix}]^T(0.3).$$

Result

$$\frac{\partial L}{\partial W_{hx}^{[1]}} = \begin{bmatrix}-0.094385167\\ -0.165174043\end{bmatrix}.$$

Layer 2 input weight gradients

$$\frac{\partial L}{\partial W_{hx}^{[2]}} = [\frac{\partial L}{\partial \hat{y}}\frac{\partial \hat{y}}{\partial z}\frac{\partial z}{\partial h_2^{[2]}}\frac{\partial h_2^{[2]}}{\partial c_2^{[2]}} \times \frac{\partial c_2^{[2]}}{\partial h_1^{[2]}}\frac{\partial h_1^{[2]}}{\partial c_1^{[2]}}]^T(h_1^{[1]})^T + [\frac{\partial L}{\partial \hat{y}}\frac{\partial \hat{y}}{\partial z}\frac{\partial z}{\partial h_2^{[2]}}\frac{\partial h_2^{[2]}}{\partial c_2^{[2]}}]^T(h_2^{[1]})^T. \tag{4}$$

Numerical substitution

$$\frac{\partial L}{\partial W_{hx}^{[2]}} = [(-1.60653066)(0.235003712)\begin{bmatrix}.5 & .5\end{bmatrix}\begin{bmatrix}1 & 0\\ 0 & 1\end{bmatrix} \times \begin{bmatrix}.75 & 1\\ .75 & 1\end{bmatrix}\begin{bmatrix}1 & 0\\ 0 & 1\end{bmatrix}]^T\begin{bmatrix}.05 & .2\end{bmatrix} + [(-1.60653066)(0.235003712)\begin{bmatrix}.5 & .5\end{bmatrix}\begin{bmatrix}1 & 0\\ 0 & 1\end{bmatrix}]^T\begin{bmatrix}.2125 & .525\end{bmatrix}.$$

Result

$$\frac{\partial L}{\partial W_{hx}^{[2]}} = \begin{bmatrix}-0.054271471 & -0.155735526\\ -0.058990729 & -0.174612559\end{bmatrix}.$$

Output weight gradient

$$\frac{\partial L}{\partial W_{hy}} = (\frac{\partial L}{\partial \hat{y}}\frac{\partial \hat{y}}{\partial z}\frac{\partial z}{\partial o})h_2^{[2]}. \tag{5}$$

$$\frac{\partial L}{\partial W_{hy}} = (-1.60653066)(0.235003712)(1)\begin{bmatrix}.63125\\ .36875\end{bmatrix}.$$

Result

$$\frac{\partial L}{\partial W_{hy}} = \begin{bmatrix}-0.238322547\\ -0.139218122\end{bmatrix}.$$

Layer 1 bias gradient

$$\frac{\partial L}{\partial \beta^{[1]}} = [\frac{\partial L}{\partial \hat{y}} \frac{\partial \hat{y}}{\partial z} \frac{\partial z}{\partial h_2^{[2]}} \frac{\partial h_2^{[2]}}{\partial c_2^{[2]}} \times (\frac{\partial c_2^{[2]}}{\partial h_1^{[2]}} \frac{\partial h_1^{[2]}}{\partial c_1^{[2]}} \frac{\partial c_1^{[2]}}{\partial h_1^{[1]}} + \frac{\partial c_2^{[2]}}{\partial h_2^{[1]}} \frac{\partial h_2^{[1]}}{\partial c_2^{[1]}} \frac{\partial c_2^{[1]}}{\partial h_1^{[1]}}) \frac{\partial h_1^{[1]}}{\partial c_1^{[1]}}]^T + [\frac{\partial L}{\partial \hat{y}} \frac{\partial \hat{y}}{\partial z} \frac{\partial z}{\partial h_2^{[2]}} \frac{\partial h_2^{[2]}}{\partial c_2^{[2]}} \frac{\partial c_2^{[2]}}{\partial h_2^{[1]}} \frac{\partial h_2^{[1]}}{\partial c_2^{[1]}}]^T. \tag{6}$$

Numerical substitution

$$\frac{\partial L}{\partial \beta^{[1]}} = [(-1.60653066)(0.235003712) \begin{bmatrix} .5 & .5 \end{bmatrix} \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \times (\begin{bmatrix} .75 & 1 \\ .75 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} .25 & .75 \\ .25 & .25 \end{bmatrix} + \begin{bmatrix} .25 & .75 \\ .25 & .25 \end{bmatrix} \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} .75 & .5 \\ .5 & 1 \end{bmatrix}) \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}]^T + [(-1.60653066)(0.235003712) \begin{bmatrix} .5 & .5 \end{bmatrix} \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} .25 & .75 \\ .25 & .25 \end{bmatrix} \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}]^T.$$

Result

$$\frac{\partial L}{\partial \beta^{[1]}} = \begin{bmatrix} -0.424733252 \\ -0.731485046 \end{bmatrix}.$$

Layer 2 bias gradient

$$\frac{\partial L}{\partial \beta^{[2]}} = [\frac{\partial L}{\partial \hat{y}} \frac{\partial \hat{y}}{\partial z} \frac{\partial z}{\partial h_2^{[2]}} \frac{\partial h_2^{[2]}}{\partial c_2^{[2]}} \times \frac{\partial c_2^{[2]}}{\partial h_1^{[2]}} \frac{\partial h_1^{[2]}}{\partial c_1^{[2]}}]^T + [\frac{\partial L}{\partial \hat{y}} \frac{\partial \hat{y}}{\partial z} \frac{\partial z}{\partial h_2^{[2]}} \frac{\partial h_2^{[2]}}{\partial c_2^{[2]}}]^T. \tag{7}$$

Numerical substitution

$$\frac{\partial L}{\partial \beta^{[2]}} = [(-1.60653066)(0.235003712) \begin{bmatrix} .5 & .5 \end{bmatrix} \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \times \begin{bmatrix} .75 & 1 \\ .75 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}]^T + [(-1.60653066)(0.235003712) \begin{bmatrix} .5 & .5 \end{bmatrix} \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}]^T.$$

Result

$$\frac{\partial L}{\partial \beta^{[2]}} = \begin{bmatrix} -0.471925836 \\ -0.566311003 \end{bmatrix}.$$

Output bias gradient

$$\frac{\partial L}{\partial q} = \frac{\partial L}{\partial \hat{y}} \frac{\partial \hat{y}}{\partial z} \frac{\partial z}{\partial q}. \tag{8}$$

$$\frac{\partial L}{\partial q} = (-1.60653066)(0.235003712)(1).$$

Result

$$\frac{\partial L}{\partial q} = -0.377540669.$$

Batch gradient update

- The worked example calculates gradients for one sample. For a mini-batch, the same forward and backward calculations are repeated for each sample using the current weights and biases. Each independent sample starts with zero initial hidden states. Contributions from both time steps and every path using a shared parameter are added within each sample.
- When the batch loss is the mean of the sample losses, each parameter's gradients are averaged across the batch. Each weight and bias is then updated once by subtracting the learning rate multiplied by its averaged gradient. The initial hidden states remain fixed at zero in this example.
- For plain mini-batch gradient descent, the update is [3]:

$$\theta_{\text{new}} = \theta_{\text{old}} - \eta \frac{\partial L_{\text{batch}}}{\partial \theta}$$

$$\frac{\partial L_{\text{batch}}}{\partial \theta} = \frac{1}{B} \sum_{s=1}^{B} \frac{\partial L^{(s)}}{\partial \theta}$$

- Here θ represents any trainable weight or bias, η is the learning rate, B is the mini-batch size, and $L^{(s)}$ is the loss for sample s. Gradients are evaluated using the current parameters before the update. For B = 1, the mean gradient equals the single sample's gradient.
- The updated parameters are used for the next batch. One pass through all training samples completes an epoch; further epochs repeat this process. Predictions on separate validation data help assess generalization.

Key takeaways

- The forward pass builds hidden states and produces a prediction; the loss measures its error. Backpropagation applies the chain rule and adds contributions from every time step and path that uses a shared parameter. The optimizer then uses the gradients to update the weights and biases after a batch.

Limitations

- The two species are processed in a chosen order to make the RNN calculations visible. This order is not a natural biological sequence and does not establish that an RNN is the best model for species-abundance vectors. The illustrative numbers provide no evidence of ASD classification performance; a real study requires suitable controls, independent evaluation and checks for study-related confounding. Longer sequences and other weight choices can produce vanishing or exploding gradients, which this two-time-step example does not demonstrate [2].

References and further reading

[1] Karpathy, A. (2015). The Unreasonable Effectiveness of Recurrent Neural Networks. A visual introduction to recurrence, shared parameters and stacked RNNs. https://karpathy.github.io/2015/05/21/rnn-effectiveness/

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[3] Goodfellow, I., Bengio, Y., and Courville, A. (2016). Deep Learning, Chapter 8, Optimization for Training Deep Models. Mini-batch gradients and gradient-descent updates. https://www.deeplearningbook.org/contents/optimization.html

[4] Liu, C., et al. (2026). GMrepo v3: a curated human gut microbiome database with expanded disease coverage and enhanced cross-dataset biomarker analysis. Nucleic Acids Research, 54(D1), D734–D742. doi:10.1093/nar/gkaf1190. https://doi.org/10.1093/nar/gkaf1190

[5] Olah, C. (2015). Understanding LSTM Networks. Further reading on the limitations of basic RNNs and the motivation for gated recurrent models. https://colah.github.io/posts/2015-08-Understanding-LSTMs/